

Business Performance Analysis

Kimia Farma - Big Data Analytics

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Disclaimer

This final project was completed as part of a project-based Business Intelligence Analyst Virtual Internship program, utilizing a structured dataset and an analytical approach supported by data visualization.

About Company



Kimia Farma is one of Indonesia's largest integrated pharmaceutical company, operating across the pharmaceutical value chain, including drug manufacturing, distribution, retail pharmacy services, clinical services, and healthcare products. With an extensive network of branches across the country, the company processes thousands of transactions every day, generating large volumes of business data.

Problem Statement

Kimia Farma operates as one of Indonesia's largest pharmacy networks, generating substantial volumes of transactional data across its branches nationwide. To support business growth and operational excellence, management requires a comprehensive understanding of business performance across different regions and over time.

Kimia Farma experienced a decline in Sales in 2021 and 2023. This decline may have been influenced by several factors, including branch and operational performance, service quality, product performance, pricing and discount strategies, customer purchasing behavior, and other external or seasonal effects.

The key business challenge is to identify the factors contributing to the Net Sales decline and identify the key areas that require attention and improvement.

The questions to analyze:

- How was the business performance? How significant the decline in in 2021 and 2023?
- Did Transaction Volume, Net Sales, and Average Sales per Transaction decline?
- Which provinces and branches contributed most to the decline?
- Which products were the biggest contributors?
- Was the decline driven by transaction volume, product performance, pricing/discounts, operational/service quality, or seasonality?

Goals

The goals of this project:

- Assess Business Performance : Evaluate the YoY changes in Transaction Volume, Net Sales, and Average Sales per Transaction.
- Identify & Evaluate Key Drivers of Sales Decline in 2021 and 2023 : Identify the provinces, branches, and products contributing most to the decline.
- Identify Priority Areas for Improvement: Determine the key areas and factors requiring the most attention based on their contribution to the decline and Provide a basis for targeted business actions to support Net Sales recovery.

Data Understanding

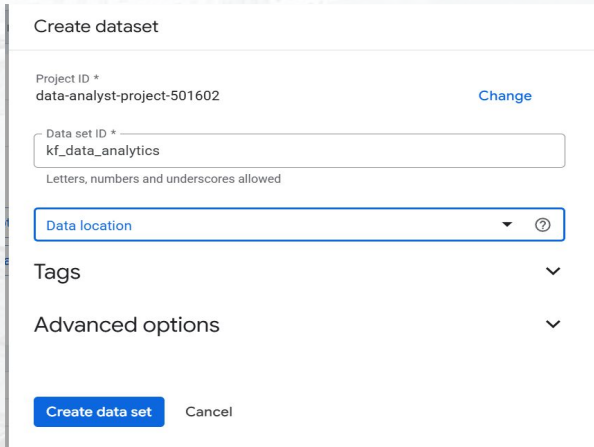
The dataset available are **kf_final_transaction** (transaction records), **kf_product** (product information), **kf_kantor_cabang** (branch information), and **kf_inventory** (inventory data). These datasets provide comprehensive information on sales, products, customer transactions, branch operations, and inventory management.

The datasets were integrated into a master analytical table using SQL joins in Google BigQuery before being analyzed with Python and visualized in Looker Studio.

- **Transaction Data:** transaction ID and transaction date
- **Branch Data:** branch, city, and province information
- **Customer Data:** customer information
- **Product Data:** product ID and product name
- **Pricing Data:** actual price and discount percentage
- **Sales & Profit Data:** net sales, net profit, and gross profit percentage
- **Service Quality Data:** branch and transaction ratings

Importing Dataset to BigQuery

The first step was importing the four CSV datasets (**kf_final_transaction**, **kf_product**, **kf_kantor_cabang**, and **kf_inventory**) into **Google BigQuery**. A dataset named **kf_data_analytics** was created, and each CSV file was uploaded as a separate table using the **Create Table** feature with **Schema Auto Detect** enabled.



Create dataset

Project ID *
data-analyst-project-501602 [Change](#)

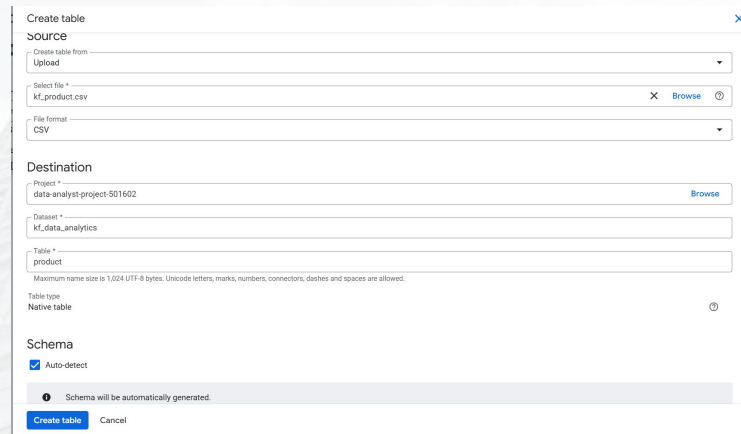
Data set ID *
kf_data_analytics
Letters, numbers and underscores allowed

Data location [?](#)

Tags [▼](#)

Advanced options [▼](#)

[Create data set](#) [Cancel](#)



Create table

Source

Create table from
Upload

Select file *
kf_product.csv [X](#) [Browse](#) [?](#)

File format
CSV

Destination

Project *
data-analyst-project-501602 [Browse](#)

Dataset *
kf_data_analytics

Table *
product
Maximum name size is 1,024 UTF-8 bytes. Unicode letters, marks, numbers, connectors, dashes and spaces are allowed.

Table type
Native table [?](#)

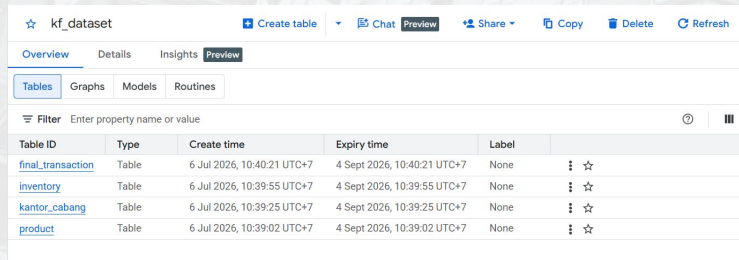
Schema

☒ Auto-detect

Schema will be automatically generated.

[Create table](#) [Cancel](#)

Successfully imported dataset to BigQuery:



☆ kf_dataset [+ Create table](#) [Chat](#) [Preview](#) [Share](#) [Copy](#) [Delete](#) [Refresh](#)

[Overview](#) [Details](#) [Insights](#) [Preview](#)

[Tables](#) [Graphs](#) [Models](#) [Routines](#)

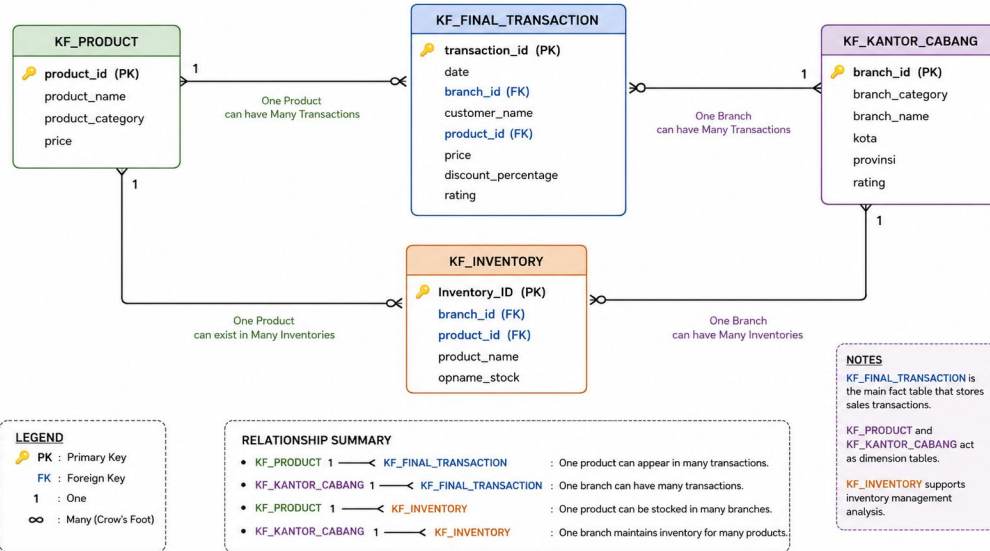
Filter Enter property name or value [?](#) [||](#)

Table ID	Type	Create time	Expiry time	Label	
final_transaction	Table	6 Jul 2026, 10:40:21 UTC+7	4 Sept 2026, 10:40:21 UTC+7	None	☆
inventory	Table	6 Jul 2026, 10:39:55 UTC+7	4 Sept 2026, 10:39:55 UTC+7	None	☆
kantor_cabang	Table	6 Jul 2026, 10:39:25 UTC+7	4 Sept 2026, 10:39:25 UTC+7	None	☆
product	Table	6 Jul 2026, 10:39:02 UTC+7	4 Sept 2026, 10:39:02 UTC+7	None	☆

Once imported, the data was ready for SQL queries, table joins, and business performance analysis.

Entity Relationship Diagram

ENTITY RELATIONSHIP DIAGRAM – KIMIA FARMA PROJECT



kf_product → kf_final_transaction: One-to-Many (1:M)

kf_kantor_cabang → kf_final_transaction: One-to-Many (1:M)

kf_product → kf_inventory: One-to-Many (1:M)

kf_kantor_cabang → kf_inventory: One-to-Many (1:M)

Analysis Table

In this project, an **analysis table** is created by aggregating and integrating data from the four previously imported datasets. The resulting table serves as the primary dataset for business performance analysis and include the following columns:

transaction_id : Unique identifier for each transaction.

date : Date when the transaction was made.

branch_id : Unique identifier for the Kimia Farma branch.

branch_name : Name of the Kimia Farma branch.

kota : City where the branch is located.

provinsi : Province where the branch is located.

rating_cabang : Customer rating of the Kimia Farma branch.

customer_name : Name of the customer who made the transaction.

product_id : Unique identifier for the product.

product_name : Name of the product.

actual_price : Original product price before discount.

discount_percentage : Discount percentage applied to the product.

gross_profit_percentage : Gross profit percentage determined based on the product price:

- Price $\leq$ IDR 50,000 $\rightarrow$ **10%**
- IDR 50,000 < Price $\leq$ IDR 100,000 $\rightarrow$ **15%**
- IDR 100,000 < Price $\leq$ IDR 300,000 $\rightarrow$ **20%**
- IDR 300,000 < Price $\leq$ IDR 500,000 $\rightarrow$ **25%**
- Price > IDR 500,000 $\rightarrow$ **30%**

net_sales : Sales value after applying the discount.

net_profit : Profit earned by Kimia Farma, calculated using the corresponding gross profit percentage.

rating_transaksi : Customer rating of the transaction.

BigQuery Syntax

```
SELECT t.transaction_id, t.date, t.branch_id, kc.branch_name, kc.kota, kc.provinsi, kc.rating AS rating_cabang, t.customer_name,
t.product_id, p.product_name, t.price AS actual_price, t.discount_percentage,
  CASE
    WHEN t.price <= 50000 THEN 0.10
    WHEN t.price <= 100000 THEN 0.15
    WHEN t.price <= 300000 THEN 0.20
    WHEN t.price <= 500000 THEN 0.25
    ELSE 0.30
  END AS gross_profit_percentage,
  t.price * (1 - t.discount_percentage) AS nett_sales,
  (t.price * (1 - t.discount_percentage)) *
  CASE
    WHEN t.price <= 50000 THEN 0.10
    WHEN t.price <= 100000 THEN 0.15
    WHEN t.price <= 300000 THEN 0.20
    WHEN t.price <= 500000 THEN 0.25
    ELSE 0.30
  END AS nett_profit,
  t.rating AS rating_transaksi
FROM `data-analyst-project-501602.kf_dataset.final_transaction` t
JOIN `data-analyst-project-501602.kf_dataset.kantor_cabang` kc
  ON t.branch_id = kc.branch_id
JOIN `data-analyst-project-501602.kf_dataset.product` p
  ON t.product_id = p.product_id;
```

Analysis Table Result

Query results [Chat](#) [View all jobs](#) [Save results](#) [Open in](#)

Job information	Results	Visualisation	JSON	Execution details	Execution graph
transaction_id	date	branch_id	branch_name	kota	provinsi
TRX5103706	2021-08-25	93529	Kimia Farma - Klinik & Apotek	Yogyakarta	DI Yogyakarta
TRX5388139	2020-12-29	24832	Kimia Farma - Klinik-Apotek-Lab...	Pekanbaru	Riau
TRX7251897	2020-02-03	20505	Kimia Farma - Apotek	Cilacap	Jawa Tengah
TRX4943675	2022-09-09	17678	Kimia Farma - Klinik & Apotek	Subang	Jawa Barat
TRX3469820	2020-06-20	28315	Kimia Farma - Klinik-Apotek Lab...	Sukabumi	Jawa Barat
TRX1213133	2021-09-17	22280	Kimia Farma - Apotek	Batam	Kepulauan Riau
TRX2020131	2020-12-16	40028	Kimia Farma - Klinik-Apotek-Lab...	Balikpapan	Kalimantan Timur
TRX5015870	2022-08-17	41343	Kimia Farma - Apotek	Semarang	Jawa Tengah
TRX7064077	2021-06-21	86546	Kimia Farma - Klinik & Apotek	Pematangsiantar	Sumatera Utara
TRX5979742	2020-12-31	18235	Kimia Farma - Klinik & Apotek	Pekanbaru	Riau
TRX2209141	2021-03-20	59571	Kimia Farma - Klinik-Apotek-Lab...	Tarakan	Kalimantan Utara
TRX5385534	2023-03-17	69280	Kimia Farma - Klinik-Apotek-Lab...	Purwakarta	Jawa Barat

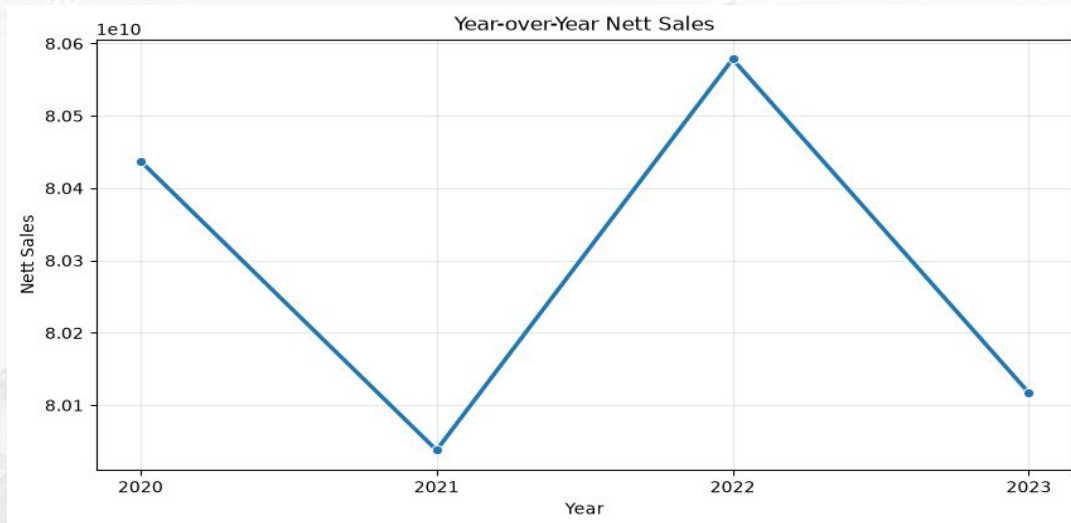
Query results [Chat](#) [View all jobs](#) [Save results](#) [Open in](#)

Job information	Results	Visualisation	JSON	Execution details	Execution graph
rating_cabang	customer_name	product_id	product_name	actual_price	discount_percent...
4.3	Derrick Wright III	KF116	Psycholeptics drugs, Hypnotics ...	251700	0.1
4.2	Elizabeth Ramos	KF116	Psycholeptics drugs, Hypnotics ...	251700	0.12
4.5	Meghan Warner	KF116	Psycholeptics drugs, Hypnotics ...	251700	0.09
4.8	Steven Roberts	KF116	Psycholeptics drugs, Hypnotics ...	251700	0.1
3.9	Linda Bruce DDS	KF116	Psycholeptics drugs, Hypnotics ...	251700	0.07
4.5	Cory Castro	KF116	Psycholeptics drugs, Hypnotics ...	251700	0.11
4.0	Stephanie Boone	KF116	Psycholeptics drugs, Hypnotics ...	251700	0.03
4.0	Mary Hughes	KF116	Psycholeptics drugs, Hypnotics ...	251700	0.03
4.5	Tamara Bruce	KF116	Psycholeptics drugs, Hypnotics ...	251700	0.04
4.8	Aaron Reed	KF116	Psycholeptics drugs, Hypnotics ...	251700	0.11
4.9	Nancy Kennedy	KF116	Psycholeptics drugs, Hypnotics ...	251700	0.1
4.0	Paul Morales	KF116	Psycholeptics drugs, Hypnotics ...	251700	0.11

Query results [Chat](#) [View all jobs](#) [Save results](#)

on	Results	Visualisation	JSON	Execution details
gross_profit_percentage	nett_sales	nett_profit	rating_transaksi	
0.2	249183.0	49836.60000000...	3.0	
0.2	226530.0	45306.0	3.0	
0.2	213945.0	42789.0	3.0	
0.2	213945.0	42789.0	3.0	
0.2	226530.0	45306.0	3.0	
0.2	216462.0	43292.4	3.0	
0.2	241632.0	48326.4	3.0	

Y-O-Y Net Sales Declining (2020-2023)



Net sales declined from 2020 to 2021, increased significantly in 2022, and declined again in 2023.

Revenue Decline Driven by Fewer Orders in 2021 & 2023

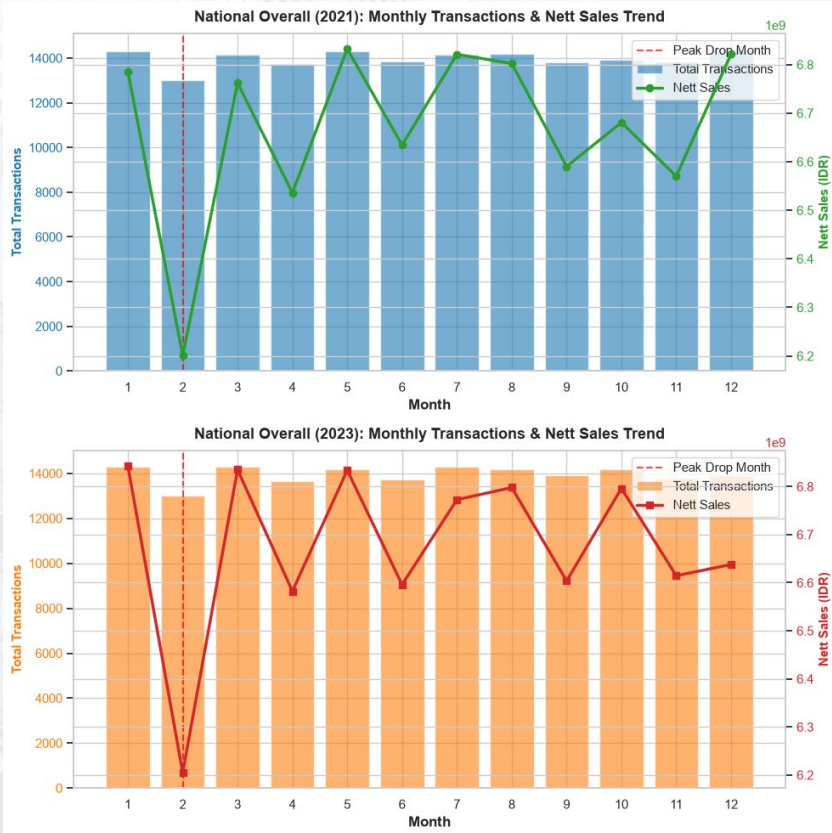
Year	Total Transactions	Nett Sales	Avg Nett Sales	% Transaction Change	% Nett Sales Change	% Avg Nett Sales Change
2020	168.651K	80.438B	476.947M	-	-	-
2021	167.697K	80.038B	477.277M	-0.57 %	-0.5 %	0.07 %
2022	168.642K	80.578B	477.808M	0.56 %	0.68 %	0.11 %
2023	167.468K	80.117B	478.404M	-0.7 %	-0.57 %	0.12 %

Volume-Driven Revenue Motion: Net Sales closely track total transaction volume. The revenue decline in 2021 and 2023 is caused by fewer completed orders, not lower order value.

Stable Purchasing Power: Average Net Sales per transaction actually increased, confirming revenue loss is not driven by lower basket sizes/lower customer purchasing power.

Volume Contraction: The business effectively maintains high-value transactions, but faces a significant drop in overall transaction volume in 2021 and 2023.

MoM Trend: February Marks the Peak Drop in Transaction Volume and Revenue in 2021 and 2023



National Performance (Jan vs Feb) :

- The monthly revenue decline from Jan to Feb from **6.79 B to 6.20 B (2021)** and **6.84 B to 6.20 B (2023)** is strictly driven by the shorter month duration (**28 vs. 31 days**, or a **9.68% reduction in operational days**).
- This structural pattern is repeat consistently across all declining years (2021 and 2023) at the national level.
- This confirms that national baseline demands stability in February.

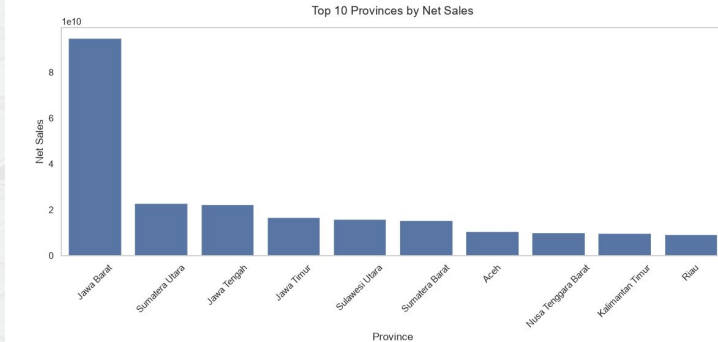
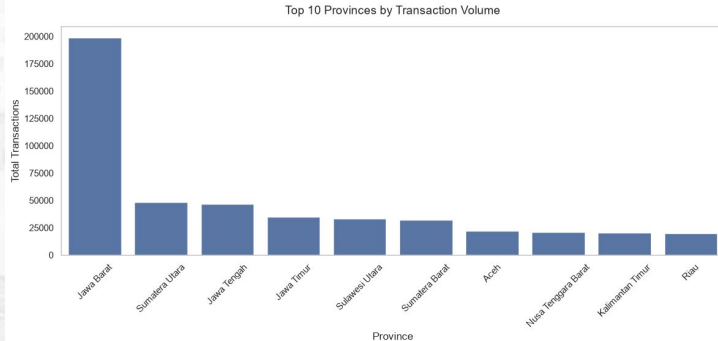
February Revenue Drop in Both Regions is Driven by Shorter Calendar Days

Year	Month	Days	Total Transactions	Daily Avg Transactions	% Daily Avg Transactions Change	Daily Avg Nett Sales	Daily Avg Transactions	% Daily Avg Transactions Change
2020	1	31	14.3 K	461.35	-	6.79 B	218.90 M	-
2021	2	28	13.03 K	465.39	+ 0.88	6.20 B	221.45 M	+ 1.17 %
2022	1	31	14.2 K	460.71	-	6.84 B	220.74 M	-
2023	2	28	13.0 K	464.64	+ 0.81	6.20 B	221.59 M	+ 0.39 %

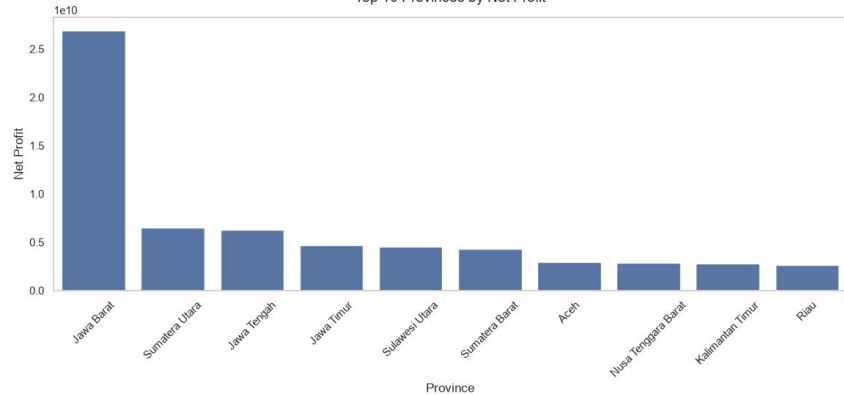
- Daily national revenue increasing from **218.90M to 221.45M** (2021) and **220.74M to 221.59M** (2023).
- **Harmonized Volume & Value:** Daily avg transactions and Daily Sales synchronously, demonstrating sustained customer purchasing power.
- **Baseline Stability:** Recurring patterns across non-leap years validate that February demand is robust.

Geographic Concentration: Core Provinces Drive the Majority of Revenue and Profitability

Top 10 Provinces with the Highest Transaction Volume, Net Sales and Net Profit



Top 10 Provinces by Net Profit



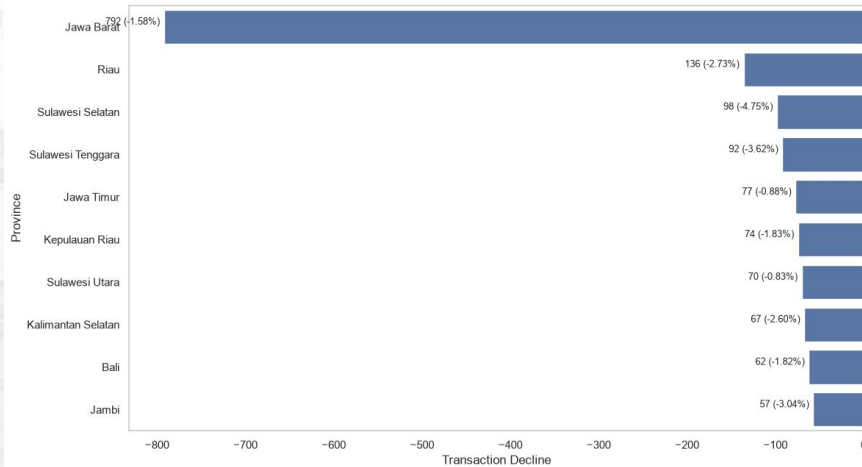
Primary Key Contributors: West Java, North Sumatra, Central Java, East Java, and North Sulawesi consistently rank as the Top 5 provinces in transaction volume, net sales, and net profit.

Secondary Key Markets: West Sumatra, Aceh, West Nusa Tenggara, East Kalimantan, and Riau complete the Top 10 high-performing regions.

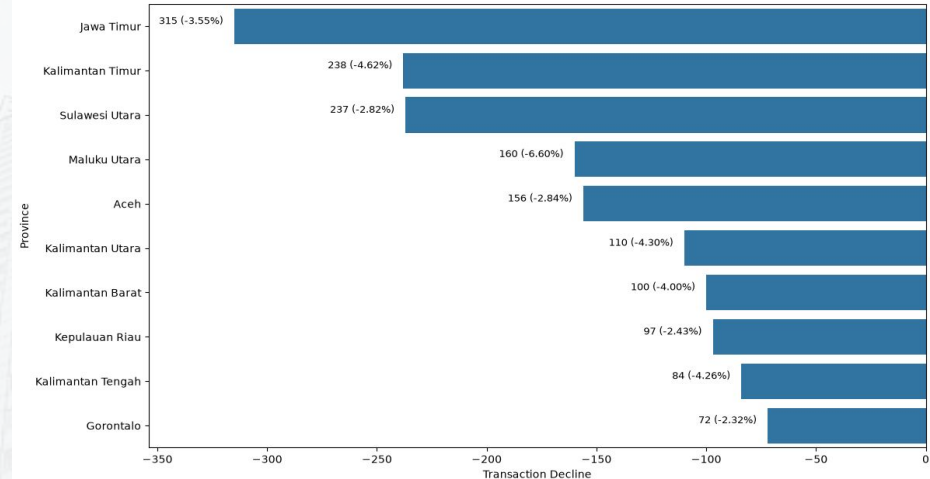
These 10 provinces serve as the primary drivers of Kimia Farma top-line revenue and profitability.

Key Provinces Driving Transaction Decline In 2021 and 2023

Top 10 Provinces with the Highest Transaction Decline (2020–2021)



Top 10 Provinces with the Highest Transaction Decline (2022–2023)



Transaction volume declines were concentrated in different provinces across 2021 and 2023. West Java recorded the largest transaction decline in 2021, while East Java and East Kalimantan recorded the largest declines in 2023.

In percentage terms, South Sulawesi saw the sharpest decline in 2021, while North Maluku, East Kalimantan, and North Kalimantan experienced the most significant relative declines in 2023.

Strategic Scope: Focus on Core / Priority Regions

West Java (2021): Highest revenue market (#1), contributed to 2021 loss.

East Java (2023): Top 5 revenue market (#4), contributed to 2023 loss.

Because these two top-contributing markets and also represent majority of volume and net sales loss, the remaining analytical scope is concentrated exclusively on **West Java (2021)** and **East Java (2023)**.

Pareto Analysis → Widespread Contraction: Revenue Drop is Systemic Across Almost All Branches, Not Specific Branches

Metric / Region	West Java (2021)	East Java (2023)
Total Underperforming Branch	269 branches out of 510 branches	51 branches out of 83 branches
Branches Driving Drop	139 branches out of 269 branches (51.7%)	27 branches (52.9%)
Total Cumulative Volume Lost	80.16%	80.03%
Diagnosed Pattern	Systemic Drop	Systemic Drop

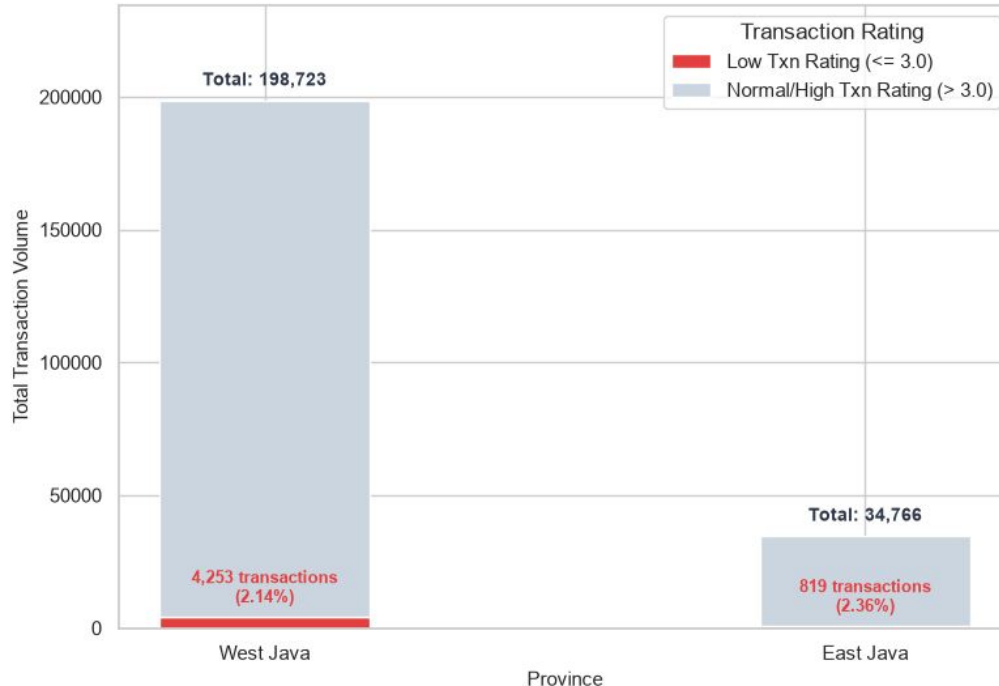
WEST JAVA in 2021: 52.7% of branches declined → 51.7% of declining branches drove ~80% of the drop

EAST JAVA in 2023: 57.3% of branches declined → 52.9% of declining branches drove ~80% of the drop

The decline is broadly distributed across the branches, showing a systemic regional issue rather than a few underperforming branches.

Low-Rated Transactions Despite High Branch Ratings

Low Transaction Rating Share in High-Rated Branches (>4.0)



Only **~2% of transactions** in West Java (2.14%) and East Java (2.36%) had **low transaction ratings (≤ 3.0)** despite **high branch ratings (> 4.0)**.

This indicates that **the majority of transactions were rated positively**, with limited evidence of widespread transaction-level service issues.

Branch and Transactions Rating Gap: Service Quality Is Not the Main Driver of Transaction Decline

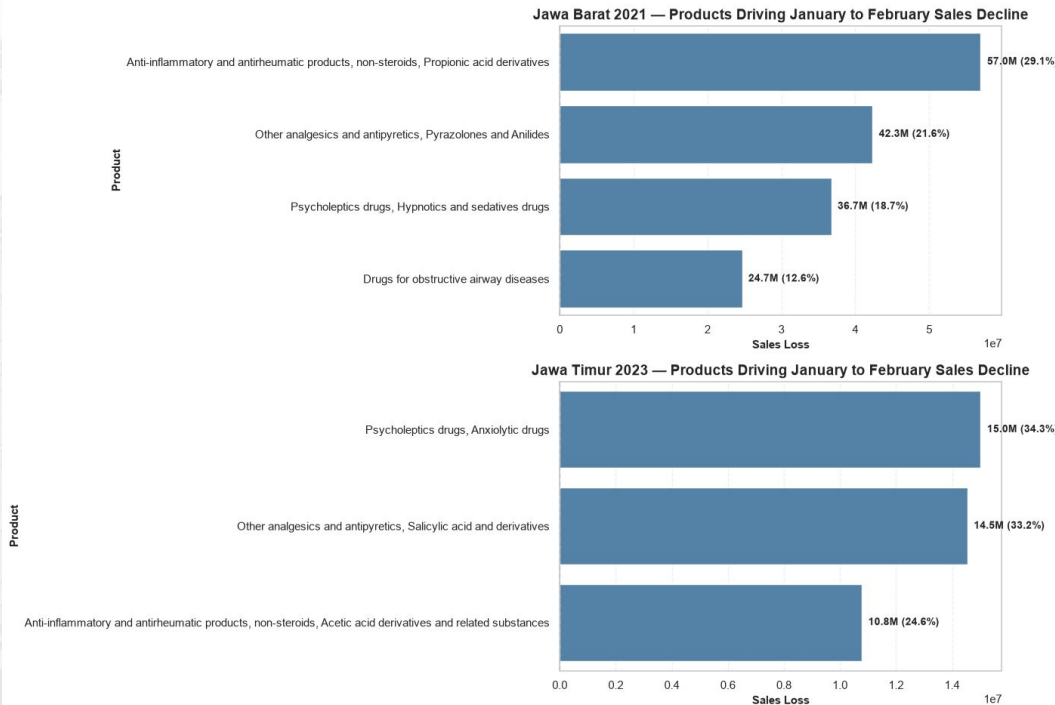
Metric / Region	West Java (2021)	East Java (2023)
Mean Branch Rating	4.45	4.50
Mean Transaction Rating	4.00	4.01
Rating Gap	0.45	0.5
Rating Gap vs. Transaction Drop Correlation	-0.055	0.058

Correlation values close to 0 indicate no meaningful relationship between Rating Gap and transaction decline.

Branch and transaction rating gaps do not show a meaningful relationship with transaction volume decline. Therefore, service quality is not a primary driver of the Net Sales decline.

Pareto Analysis → Sales Decline in West Java (2021) and East Java (2023) Is Highly Concentrated in a Few Product Categories

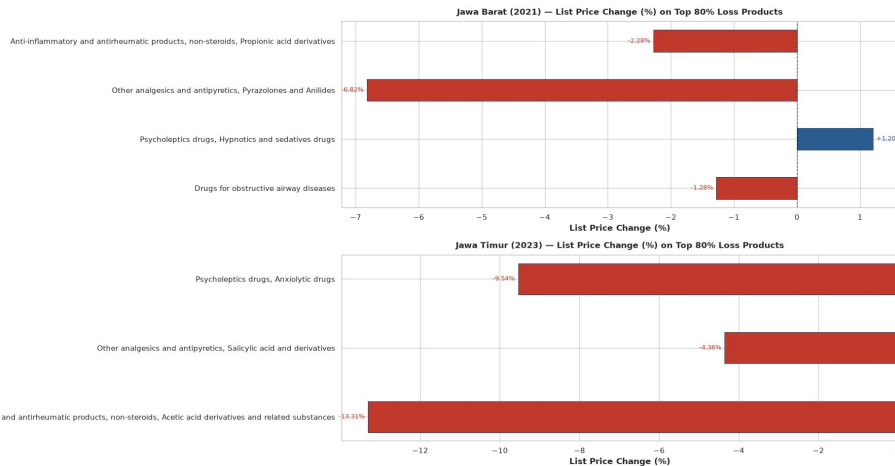
Products Contributing as the Main Drivers of January to February Sales Decline in 2021 and 2023



- 81.99% of West Java's 2021 sales loss came from just 4 product categories
- 92.18% of East Java's 2023 loss came from only 3 categories.

Broad Price Reductions Failed to Stimulate Demand / Drive Volume UpLift

List Price Changes on Top 80% Loss Products (Jan vs Feb)



Asymmetric List Price Adjustments:

West Java (2021):

- List price adjustments ranged from **-6.82% to +1.20%** (largest drop: *Analgesics & Antipyretics* at **-6.82%**).
- Price Hike Anomaly:** *Psycholeptics, Hypnotics & Sedatives* was the only product which list price increase (**+1.20%**).

East Java (2023):

- List price reductions were significantly more aggressive across all top loss contributors, reaching up to **-13.31%** (*Anti-inflammatory products*).

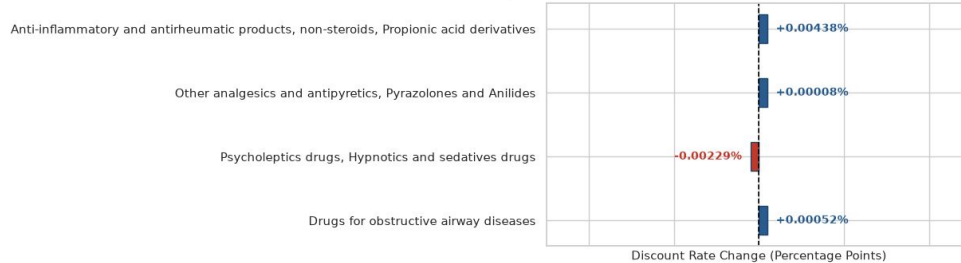
Two-Way Commercial Failure:

- Failed Volume Uplift (Price Cuts):** Lowering list prices failed to stimulate sufficient order volume under low price-sensitivity conditions, directly declining sales.
- Volume Contraction (Price Hikes):** Raising list prices triggered sharp drops in order frequency, driving revenue loss.

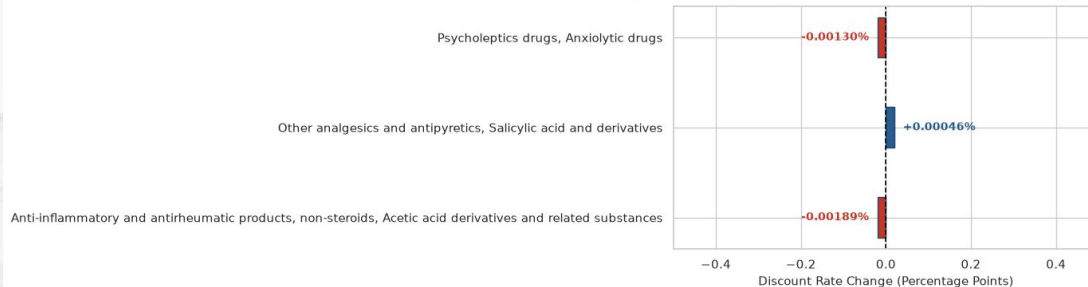
Stable Discount Rates Confirm Loss Was Driven by List Price Cuts

Discount Rate Fluctuation (Jan vs Feb) on Top 80% Loss Products

Jawa Barat (2021) — Discount Rate Delta (Percentage Points)



Jawa Timur (2023) — Discount Rate Delta (Percentage Points)



Flat Discount Rates: Discount rate fluctuations (Jan vs. Feb) across top 80% loss-contributing products remained virtually flat, ranging within a negligible range of **-0.0023% to +0.0044%**.

No Promotional Leakage: Net Sales decline were conclusively **not driven by excessive trade promotions or discount spillage**, but rather strictly by catalog list-price adjustments.

Conclusion

Key Findings:

1. Volume-Driven Sales Decline (Stable Basket Size)

- The decline in Net Sales is purely a **transaction volume problem**. Average Order Value (AOV) remained stable or increased, confirming that customers are ordering less frequently, not spending less per order.

2. Operational & Service Quality Excluded from the main driver of Sales Decline

- **Calendar Impact:** February's monthly dip is an artifact of fewer operating days (28 vs. 31 days). Daily transaction run-rates actually increased.
- **Systemic Distribution:** Over 50% of branches contributed to losses, ruling out individual branch underperformance. Branch ratings remained high (>4.0) with zero correlation to sales drops.

Conclusion

Key Findings :

3. High Concentration in Key Products

- Sales leakage is not broad-based. It is heavily concentrated in **3-4 core high-revenue categories**: *Psycholeptics, Analgesics & Antipyretics, Anti-inflammatory, and Obstructive Airway drugs.*

4. Two-Way Pricing Failure:

Unstrategic price changes failed on both sides, while promotional discounts stayed virtually unchanged:

- **Failed Price Cuts:** Broad price reductions (up to **-13.31%**) failed to stimulate volume demand, directly declining Net Sales.
- **Price Hike Anomaly:** An isolated price increase (**+1.20%** on *Hypnotics & Sedatives* in West Java) triggered a severe transaction drop.

Key Takeaway

- **Core Commercial Root Cause: Volume Drop & Pricing Misalignment**

Analysis confirms that **Net Sales decline** is **fundamentally driven by a severe drop in transaction volume, heavily concentrated in key products** West Java across (2021) and East Java (2023) and **compounded by unstrategic catalog pricing adjustments** (Two-Way Pricing Failure).

- **Operational & Service Quality Excluded**

Service quality, branch performance, and calendar effects (February) are ruled out as root causes. The issue is purely commercial and market-facing.

- **Demand Contraction Requires Field Market Intelligence**

While internal logs confirm a true transaction drop (stable AOV with lower order frequency), determining the exact driver of external demand contraction—such as competitor price wars, pharmacy credit limits, or stockouts—requires **field market intelligence** and operational validation.

Actionable Commercial Recovery Strategies

Strategic Actions to Rebuild Order Volume and Protect Net Sales

1. Prioritize Core Loss-Contributing Products

- Focus commercial resources directly on high-revenue SKUs experiencing severe transaction drops (*Psycholeptics, Analgesics, Anti-inflammatory, Airway drugs*) to maximize recovery impact.

2. Establish Disciplined Catalog Pricing (Cuts & Hikes)

- **Stop Unnecessary Price Cuts:** Set a firm price on low-sensitivity products to prevent Net Sales erosion.
- **Re-evaluate Price Hikes:** Review unmanaged price increases (e.g., *Hypnotics & Sedatives* +1.20%) to prevent further order contraction.

Actionable Commercial Recovery Strategies

Strategic Actions to Rebuild Order Volume and Protect Net Sales :

3. Shift to Targeted Commercial Promotions and Incentives

Replace permanent list-price cuts with flexible, performance-driven trade promotions:

- **Volume-Based Tiered Discounts:** Higher discounts for larger order volumes (*basket sizes*).
- **Strategic Product Bundling:** Pair high-demand anchor products with declining SKUs.
- **Pharmacy & Healthcare Facility Incentives:** Extend payment terms (TOP) and offer display support.
- **Performance Channel Rebates:** Post-period credit notes for sub-distributors hitting quarterly targets.

4. Monitor Elasticity & Regional Priorities

- Regularly track price changes against volume response and Net Sales.
- Deploy regional recovery plans and set clear volume targets in **West Java** (2021 priorities) and **East Java** (2023 priorities).

5. Limited Data

- While the current transaction dataset is robust for isolating commercial root causes, further operational validation is constrained by the absence of Customer Master Data and historical Branch Stock Opname/Out-of-Stock logs.

Link Project & LinkedIn

<https://github.com/grace-nc/kimiafarma-business-performance-analysis>

<https://datastudio.google.com/reporting/162035e9-b789-43eb-8c4e-bb0dde04e1a7>

<https://www.linkedin.com/in/grace-natalie-catherine>

Thank You



Rakamin
Academy



kimia farma

The logo for Kimia Farma, featuring a white arc with a dotted pattern above the company name in a bold, italicized sans-serif font.